

ROS Backend Challenge

Introduction

The primary goal of this challenge is to develop an autonomous mobile robot system using ROS2 Jazzy in a Gazebo simulation.

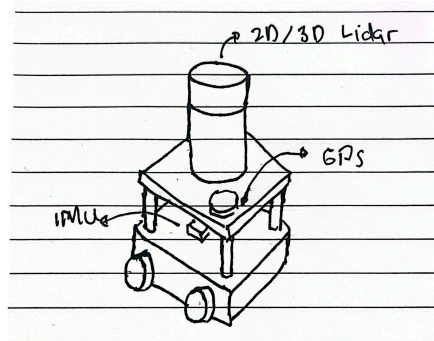
Component	Requirement
ROS Version	ROS2 Jazzy Jalisco
Simulator	Gazebo Harmonic
Robot Model	Turtlebot3 Waffle, or a custom URDF/SDF model (preferred).
UI Framework	Any (RViz2 plugin, web-based, Qt, etc.)

Task I - Robot and Simulation Setup

Set up the following Gazebo simulation environment:

- Arena size: 10 m x 10 m enclosed with walls on all sides
- Obstacles: 3 static cylinders with 0.5m diameter (place anywhere)
- The robot can be spawned anywhere inside the arena

You can use available models such as the Turtlebot3 Robot or make your own custom URDF model for your robot (preferred). Use this design as reference with arbitrary reasonable dimensions.



Task 2 – SLAM and Map Building

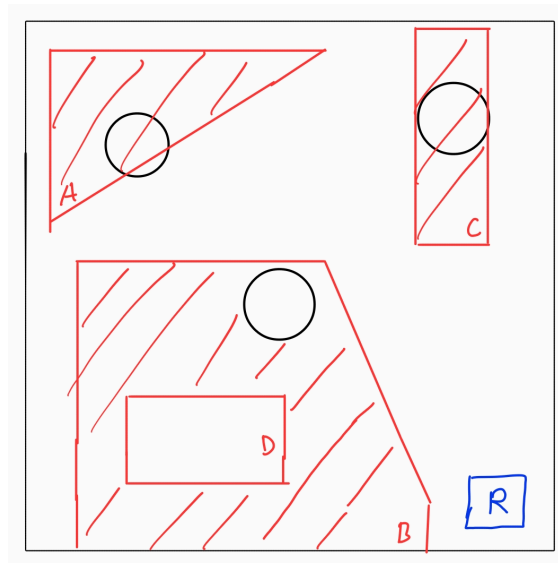
Create a system where the user can build the map of the arena. You can make the system using teleoperation to go around the arena. Optionally, you can also make this process to be automated with automatic map exploration packages.

Task 3 – Area Coverage System

Using the map from Task 2, build a system that allows the user to define a coverage area on the map. The area can be any arbitrary shape.

For example, on the map, the user defined area A, B, and C to cover.

- Check if the area is invalid (lets say the area is impossible to cover because it is enclosed by a wall)
- An area that overlaps with obstacles should be valid, but the system will exclude the obstacle as the area it covers.
- (Optional) Users can exclude an area. In this case, area D is excluded from the coverage area.



After the user defines the area to cover, the robot (R) should start the **boustrophedon movement** on those areas.

- In Area A, B, and C, the robot will cover the whole area while also avoiding the obstacles.
- (Optional) Make the option so that the order of the area of which the robot will cover can be rearranged to be more optimal. (Example: the robot covers area B first, then area A, then area C, because B is the nearest to the robot).

Documentation and Submission

1. Store the code on GitHub repository and **share the GitHub repository when you start doing the project** to wa.me/+6287818152770. It must be in a public repository.
2. Please **make the documentation for the project in the README.md** (you can make many pages too). Make sure the documentation contains:
 - Project Overview
 - How to Setup
 - How to Use (with configuration guide)

- The System Architecture
 - (Add more if needed)
3. Create a maximum **10 minutes video using English** containing the demonstration and explanation of how you implemented it, the limitations, and what needs to be improved.
 4. **After finishing the video**, please **send the video (must) and other related files (if needed) to the email itbdelabo@gmail.com** with subject: **Robotics Case Study Submission**
***Note:** If you are working on other questions (Electrical Design and/or ROS - Web Integration), please attach them together in the same email.*
 5. The video is also to be shown during the interview and please explain during the interview.